Analogical Mapping Examples

In **Analogical Mapping** (the theory of structural alignment formulated by psychologist Dedre Gentner), "figures" are not placed in analogy because they look physically alike, but because **the internal relationships between components in the *Source* (Base) system are identical to the internal relationships in the *Target* system**.

The three classic examples studied in cognitive psychology highlight the alignment of these figures:

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1. The Solar System (Source) and Rutherford's Atom (Target)

This is the most famous academic example used to explain relational mapping.

- **The Analogous Figures:** The **Sun** is analogous to the **Nucleus**, while the **Planet** is analogous to the **Electron**.
- **The Mapped Properties:** The mind ignores physical properties (the Sun is hot and yellow, whereas the nucleus is not). The mapping occurs strictly on **relational properties**:
  - *Sun (more massive) → attracts → Planet (less massive)*
  - *Nucleus (more massive) → attracts → Electron (less massive)*
  - *The Planet **orbits around** the Sun → The Electron **orbits around** the Nucleus*.

2. Water Flow (Source) and Heat Flow or Electricity (Target)

This mapping is constantly used to explain abstract physics concepts to students who are unfamiliar with them.

- **The Analogous Figures:** The **Pipe** is analogous to the **Copper Wire** (conductor); the **Water** is analogous to the **Electric Current** or **Heat**; the **Water Pressure** is analogous to **Voltage** (potential difference).

- **The Mapped Properties:**
  - Water moves from a point of higher pressure to a point of lower pressure → Current moves from a point of higher potential to a point of lower potential.
  - The narrower the pipe, the less water passes through (hydraulic resistance) → The thinner the wire, the less current passes through (electrical resistance).

3. The Military Attack Problem (Source) and the Tumor (Target)

Taken from the famous psychological experiment by Duncker and adapted by Gick and Holyoak to study analogical problem solving.

- **The Source Situation:** A general wants to capture a fortress in the center of a country. There are many roads leading to it, but they are mined: if an entire army passes, they will detonate. Solution: the general **divides the army into small groups** and sends them simultaneously down different roads, converging on the fortress at the same time.
- **The Target Situation:** A doctor needs to destroy an inoperable malignant tumor using a radiotherapy beam. If the beam is too powerful, it will destroy the surrounding healthy tissue.
- **The Analogous Figures and Mapping:**
  - The **Fortress** is analogous to the **Tumor**.
  - The **Army** is analogous to the **Radiotherapy Beam**.
  - The **Mined Roads** are analogous to the **Healthy Tissue**.
- **Resolution by Analogy:** The brain maps the structure and projects the solution (*Candidate Inference*): the doctor **divides the beam into multiple low-intensity rays** coming from different directions, which all converge simultaneously on the tumor without damaging the healthy tissue they pass through.

Visual Summary of the Mapping

Structural Role	Source Domain (Base)	Target Domain (Target)
Central Element	Sun	Nucleus
Orbiting Element	Planet	Electron
Binding Force	Gravity	Electromagnetic Attraction

In the brain, analogical mapping functions like a "copy and paste" not of the objects themselves, but of the geometric, logical, or relational rules that govern them.